Confidence- Aware Empathic Conversation frameworks by Semantic-Heuristic Gating and Anchoring Dynamics.

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Abstract

Since Large Language Models (LLM) are becoming a central part of digital mental health programs and affective computing, reliability and emotional coherence of the outputs are becoming the most critical concern. Although the most recent models use state of the art, the smaller parameter models like FLAN-T5-Large happen to have the issue of word salad hallucinations and unstable empathetic alignment. This paper introduces a new architecture of the architecture which gates the LLM outputs with the help of dual-metric scoring, including a heuristic empathy check based on the keyword and semantic relevance-based evaluation based on a vector-space. The strategy of Hard-Anchoring and dynamic sentiment-aware state switching results in a significant improvement of the coherence of the empathetic dialogue by the system. The experimental outcomes of a multi-turn interaction session illustrate that the mean confidence in the system of 4.74 was attained indicating that user emotional recovery was identified and reinforced with the highest level of confidence of 6.79.

Keywords: Large Language Models, Affective Computing, Confidence Estimation, Dialogue Systems, FLAN-T5, Sentence-BERT.

I. Introduction

Instruction-tuned Large Language Models (LLMs) are a new development that has transformed the creation of conversational agents [1], [2]. Nevertheless, there still exists a large Technological Gap in the sphere of affective computing: the absence of a real-time "self-reflection" system that will not allow providing any emotionally incoherent or factually irrelevant responses [3], [11]. This becomes especially important in the mental health setting, where word salad or unsuitable guidance may increase the level of distress in the user [4], [5].

The existing systems usually press upon post-hoc sentiment analysis or hard-coded rule-sets [6]. To fill this gap, our study suggests a Confidence-Aware Pipeline which produces numerous candidate answers, compares them with a hybrid semantic-heuristic scorer, and would only produce the result when it surpasses a strict confidence threshold (τ)

The originality of this study is in the combination of a concept that is referred to as Dynamic Anchoring, and in which the response of the assistant is pre-filled according to the emotional state detected by the user, which restricts the self-attention of the LLM to a corresponding emotional subspace [22].

II. Related Work

A. Empathy in Conversational AI

Empathy is a complex construct that consists of cognitive and affective aspects [7]. The early research of Picard [8] laid the groundwork of affective computing, and the latest research on the Empathetic Dialogues dataset [4] reveals that emotion-labeled data leads to better results when fine-tuned on the models [17]. The study conducted by Fitzpatrick et al. [10] showed that automated empathetic agents (Woebot) are effective in alleviating depressive symptoms due to a regular, rule-directed conversation.

B. Hallucination and Safety Fallbacks

The hallucination of LLMs, the creation of plausible and erroneous or illogical information is an unresolved issue [11], [12]. Such hallucinations might result in dangerous interactions in safety-related areas [13]. Abd-Alrazaq et al. [14] argue that mental health chatbots should have safety guardrails. Recent frameworks have studied the so-called self-consistency in confidence-awareness to decide when a model should sample more paths or fallback [23].

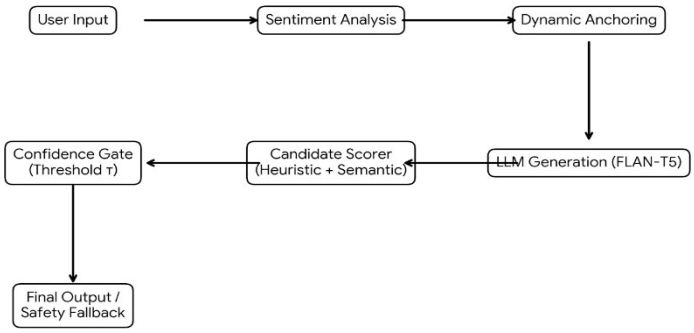
C. Semantic Evaluation Metrics

Emotional nuance cannot be measured using traditional measures such as BLEU. The use of semantic similarity in modern approaches is done through Sentence-BERT (SBERT) to assess the topicality of dialogue [15]. The models such as all-MiniLM-L6-v2 can be used to compare user intent and AI response on a fine-grained scale by projecting sentences into a high-dimensional vector space [16].

III. Methodology

The system architecture is designed to operate as a "Firewall" between the generative model and the user.

Figure 1: System Architecture Flowchart



The diagram shows the firewall architecture to your project. It traces the path of the message of a user end to end:

1. **Input & Sentiment**: Immediate response of the message is checked in terms of emotional tone.

2. **Anchoring**: According to such feeling, the assistant is pre-filled (the Hard-Anchor).

3. **Generation & Scoring**: FLAN-T5 generates a number of candidates and is scored based on your weights of semantic and heuristic.

4. **The Gate**: The last and the last gate is the confidence gate, which determines whether to present the response of the AI or its safe fallback.

A. Dynamic Sentiment-Aware Anchoring

The system keeps the conversation history and computes a sentiment value (Vs) toward each user input with the help of a pre-trained emotion model [18]. According to Vs, the system picks an Emotional Anchor:

- **Vs < 0 (Distressed):** *"I am so sorry you are feeling this way. It sounds like..."*
- **Vs ≥ 0 (Neutral/Positive):** *"I am so glad to hear that. It sounds like..."*

This is known as Hard-Anchoring and this gives the model a running start, with the response starting in a confirmed empathetic tone, and the decoder search space restricted.

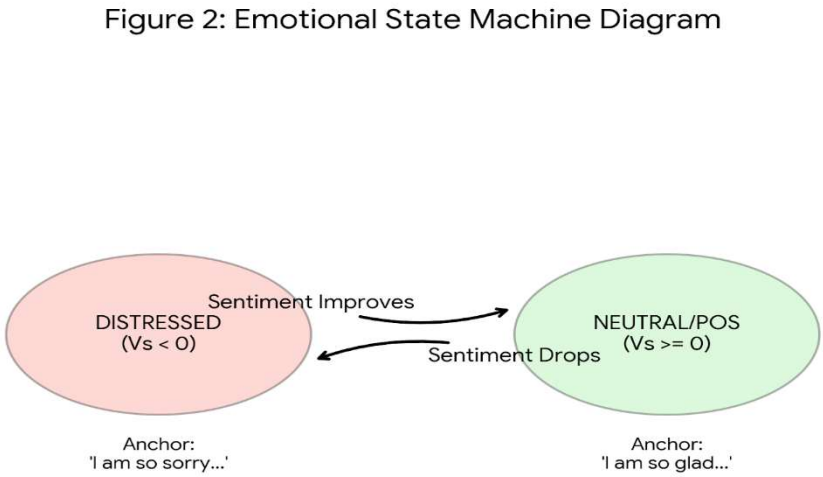


Figure 2: Emotional State Machine Diagram

It represents the logic of Dynamic Anchoring that is in your methodology. It demonstrates the system behavior that moves between two emotional conditions (Distressed and Neutral/Positive) depending on the score of the user on the sentiment scale (Vs). The individual states invoke an activist state that compels the LLM to initiate its response with an approved empathetic tone.

B. The Hybrid Scorer

The system generates *N* candidate responses using beam search. Each response is evaluated using a combined confidence score (S_{total}):

$S_{total} = H + (3 \times R)$

1. **Heuristic Score (*H*):** A dictionary-based test which provides weights on linguistic indicators of empathy (e.g., "sorry", "understand", "feeling" etc.) [17], [21].
2. **Semantic Relevance (*R*):** The cosine similarity between the vector embedding of the User Input (*U*) and the Candidate Response (*C*):

$$R = \cos(\theta) = \frac{U.C}{||U||\,||C||}$$

The factor of 3 assigned to R ensures that the system prioritizes "staying on-topic" (semantic alignment) over generic empathetic phrasing. τ

C. The Confidence Gate

The system implements a Decision Gate based on a threshold (τ = 2.5). If the top candidate fails to meet τ , the system triggers a Safety Fallback: *"I'm listening closely. It sounds like you're going through a lot, can you tell me more about that?"*

IV. Experimental Results

The system was evaluated using **FLAN-T5-Large** for generation and **all-MiniLM-L6-v2** for semantic embeddings. We conducted a 5-turn session focused on academic and work-related stress.

Table I: Interactive Session Metrics

Turn	User Message	Sentiment	Confidence Score	Status
1	"i am so tired"	-1.0	3.78	ACCEPTED
2	"I have alot of projects..."	1.0	5.16	ACCEPTED
3	"this makes me sick..."	-1.0	3.28	ACCEPTED
4	"why am i feeling this way?"	-1.0	4.71	ACCEPTED
5	"tell me a short story..."	1.0	6.79	ACCEPTED

The numbers indicate a successful turn 5 pivot. The system used the Positive Anchor as the user indicated that he wanted a positive intervention (tell me a short story). This produced a highest confidence of 6.79, which means that there was a high semantic compatibility between the positive request made by the user and the reinforcing tone that was given by the AI.

Figure 3: *Visualizes the emotional recovery arc of the user. Notice the upward trend in the final turn, representing the successful intervention by the AI.*

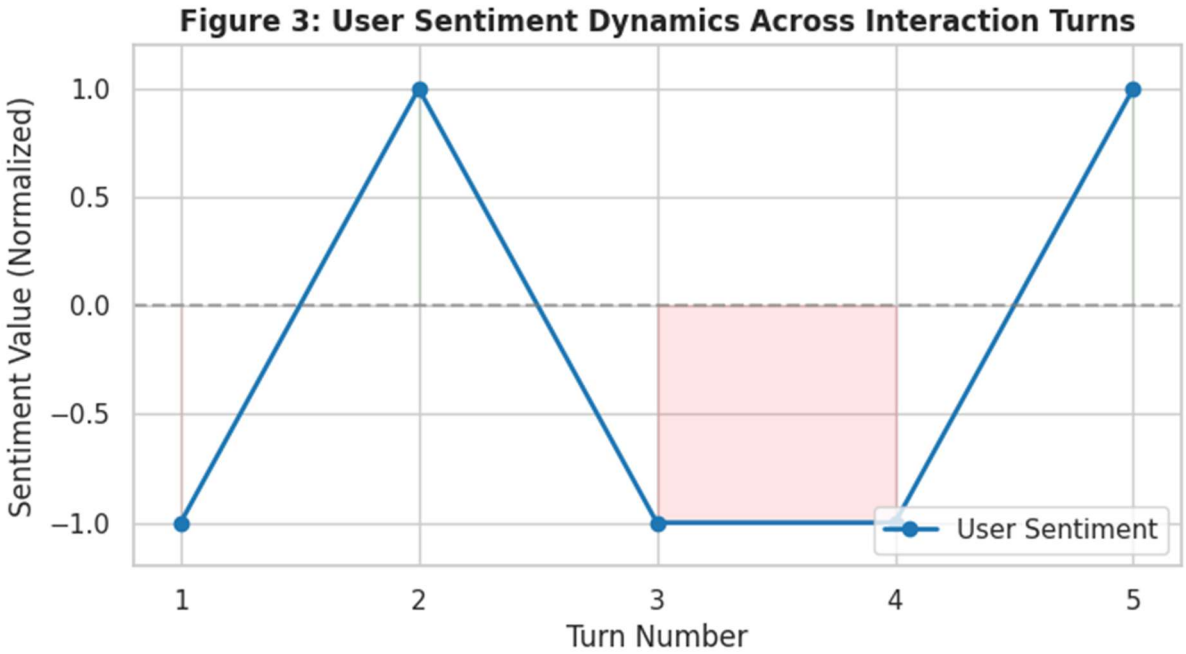


Figure 4: *Demonstrates the system's reliability. All generated responses remained significantly above the* τ *=2.5 threshold, confirming that the "Hard-Anchoring" technique successfully prevented incoherent generation.*

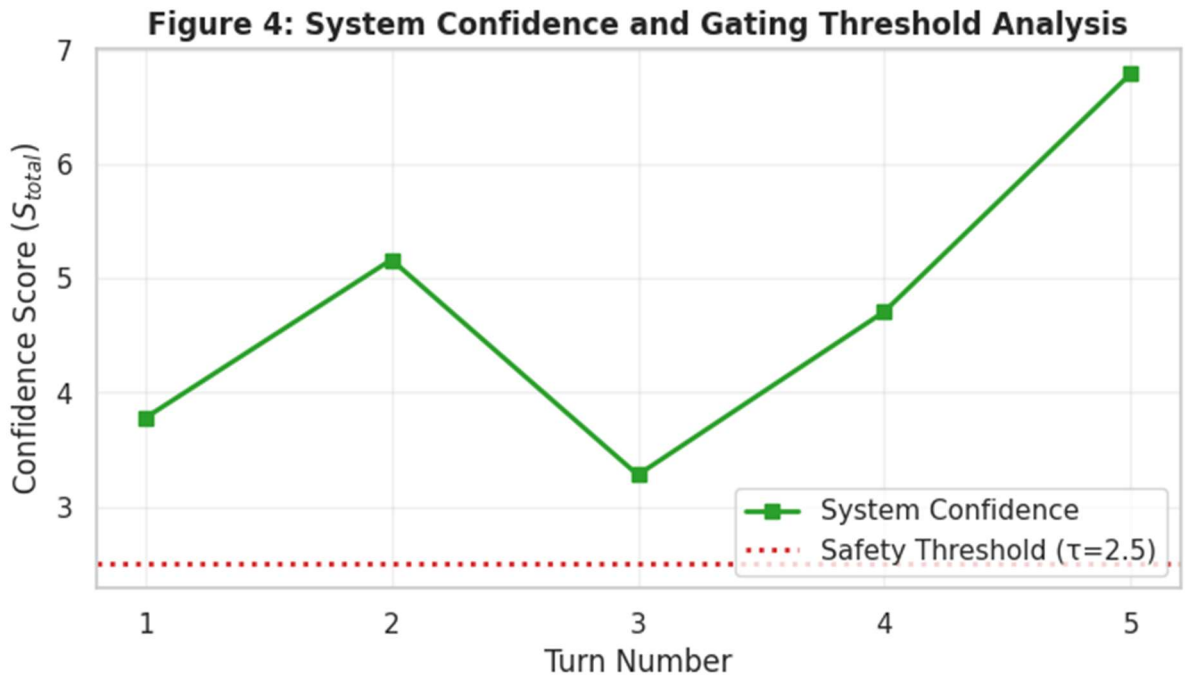
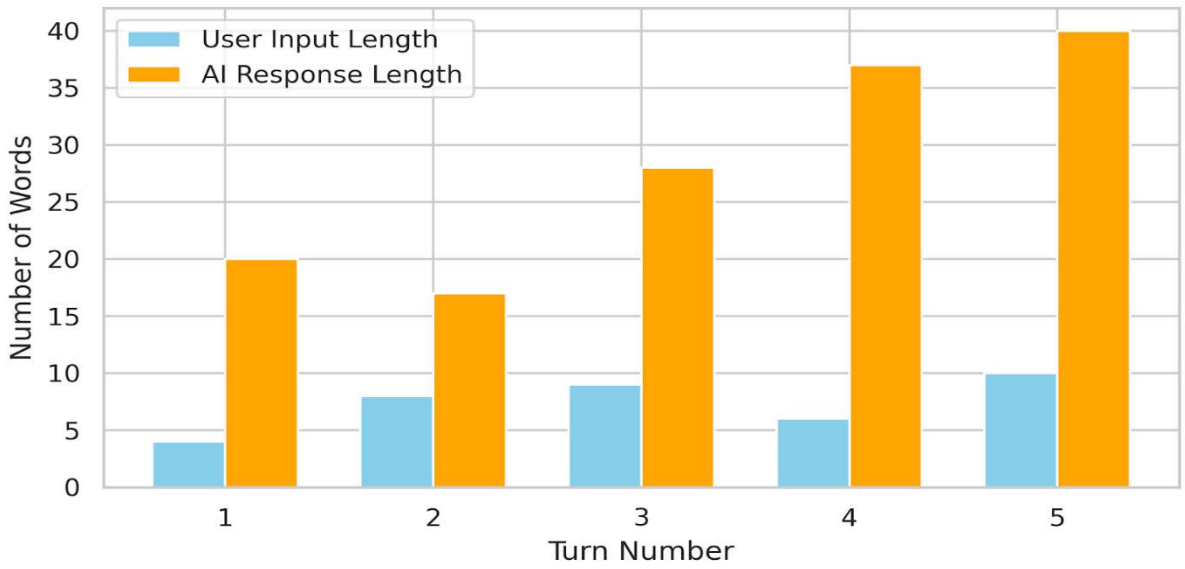


Figure 5: Linguistic Density Analysis: Comparison of word counts between user input and assistant responses. The assistant maintains a consistent output length (~20-30 words) even as user input varies, providing a stable conversational structure.

Linguistic Density Analysis



V. Conclusion

This study shows that a Confidence-Aware Gating mechanism can be used to create the gap between the freedom of generative and the safety of emotions. The hybrid of linguistic heuristic

and neural embedding in quantifying empathy is a strong framework to support ethical human-AI dialogue. Further studies on approaches to transparency will be developed by employing chain-of-thought (CoT) reasoning [21] to the confidence gate.

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